

Open Design Project: Spotted Lanterflies

Team: *Sinicus* **Client(s):** *Cornell CALS Extension / E&J Gallo Winery / National Grape*

Problem statement

SLFs feed on the phloem of grapevines, which reduces nitrogen and carbon in the roots and soil and affects the taste and size of future grape harvests. If SLFs are left unchecked, they can ultimately kill an entire vineyard. Moreover, the accepted tolerance of SLFs in grape products is low, with only 1–2 SLFs per kilogram of grapes required to reject a 22 ton batch. Ensuring that there are no SLFs on the vine before harvest begins will also address the contamination problem without adding complexity or delaying the harvesting process.

Impact

Solving this problem is beneficial to the grape industry, as preventing SLFs from accessing grapevines decreases the number of throwaway batches while improving the quality of present and future harvests alike. Grape harvests have an extremely low tolerance for the number of SLFs allowed. Just 1–2 grams of SLFs (about 1–2 SLFs) per kilogram of grapes is sufficient to throw out the entire 22-ton batch, leading to loss of profit and wasted material. Eliminating SLFs before they ever reach the vine preserves the integrity of the current crop of grapes, improving the profitability of harvest. Furthermore, SLFs feed on grapevine phloem, and while healthy vines can produce quality fruit for 50+ years, this useful life diminishes as SLFs cause vine health to deteriorate. Eliminating lanternflies will preserve the health of the vines, prolonging the useful life of the vines and ensuring profitability on a longer time scale.

Proposed direction

(primary): <1-line name> **Electrical Fence Lure and Pesticide Spraying Mechanism** <1–2 sentences.> **How it would be used:** <2–4 bullet “user flow” steps.> **Why it’s better than the status quo:** <2–3 bullets.> **End-of-semester proof-of-concept:**

Key Risks / Unknowns

The key risks of focusing on our proposed problem are the scale of the solution required. New York State has over 30,000 acres of vineyards, and to block SLFs from entering vineyards would require many resources just to physically cover a lot of space. In addition, because our solution focuses on prevention, if it is ineffective and does allow a significant number of SLFs to enter vineyards, then we have no way to sort SLFs from grapes during harvesting.

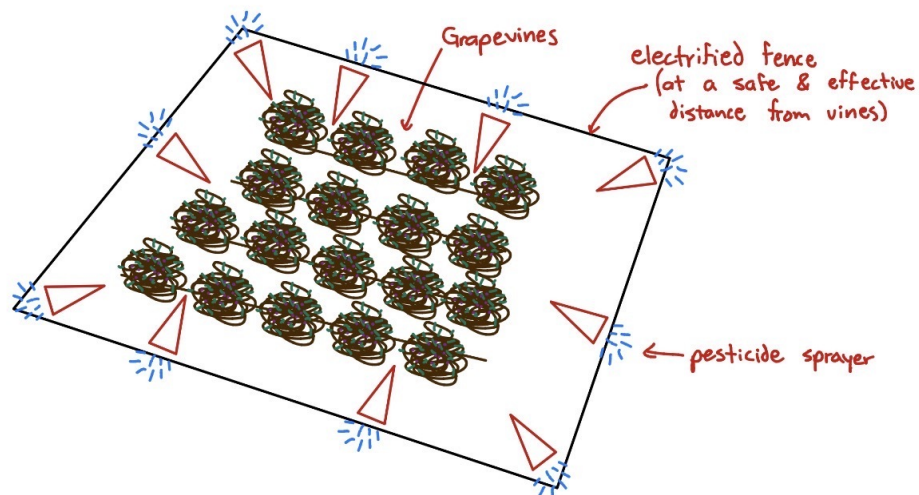
Furthermore, the use of an electrified wire to attract spotted lanternflies may cause ecological disturbance beyond the desired scope of impact. More specifically, it is possible that other insects (perhaps even those beneficial to agriculture such as pollinators) may be attracted to the utilized frequencies and would also be eliminated by the pesticide sprayer. Additionally, depending on the particular implementation of this solution, it is possible that this electrified wire would pose harm to larger animals passing through areas adjacent to grapevines, such as deer, raccoons, foxes, rabbits, and other mammals commonly found in the Northeastern U.S.

Questions for the client

1. Has anyone tried to attract SLF's before?
2. When spotted lanternflies do get on the grapevines, do they jump around on the same vine, or jump to different grapevine rows?

Figure

Full Scale Solution



MAE 2250 Proof of Concept

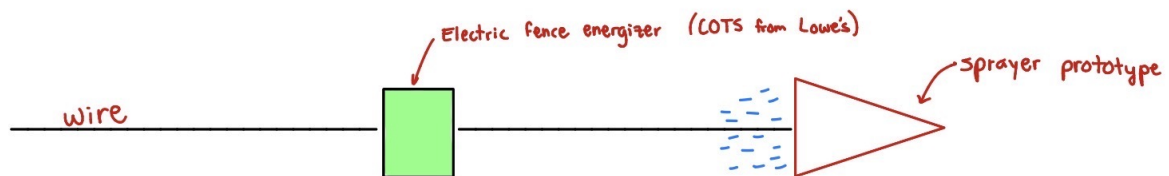


Figure 1: Diagram of intended full scale solution and proof of concept for this class.

References

- Harner, Andrew D., Heather L. Leach, Lauren Briggs, and Michela Centinari. 2022. "Prolonged Phloem Feeding by the Spotted Lanternfly, an Invasive Planthopper, Alters Resource Allocation and Inhibits Gas Exchange in Grapevines." *Plant Direct* 6 (10). <https://doi.org/10.1002/pld3.452>.
- Petit, Elsa, Zoe Robinson, Jessica Ellis, Max Resnick, Amber Ali, and Sonia Schloemann. n.d. "2021 New England and New York Grape Production Survey." *Fruit Notes* 88: 2023. Accessed February 12, 2026. <http://umassfruitnotes.com/v88n1/FN6.pdf>.
- Rohde, Barukh, Miriam F Cooperband, Isaiah Canlas, and Richard W Mankin. 2022. "Evidence of Receptivity to Vibroacoustic Stimuli in the Spotted Lanternfly *Lycorma Delicatula* (Hemiptera: Fulgoridae)." *Journal of Economic Entomology* 115 (6): 2116–20. <https://doi.org/10.1093/jee/toac167>.